Saat

Hourly proof for clean energy: AI-verified hourly certificates, traded peer to peer, retired on-chain only against same-hour load

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Track: Renewable Energy & Energy Trading
github.com/yy5652-hash/saat
Demo: yy5652-hash.github.io/saat
Live on Ethereum Sepolia: 0x616F…2F4A

The problem

Companies call themselves "100% renewable" because they buy certificates equal to their **annual** electricity use. Electricity is consumed **hour by hour**, and a solar MWh from noon cannot have powered a hotel at 9 p.m. On **Türkiye's real 2025 grid** (EPİAŞ hourly generation via the IEA Real-Time Electricity Tracker), a 24/7 consumer holding wind and solar certificates for 100% of its annual use is matched only **72.3%** hour by hour, and 128 kg CO_2e/MWh is reported as zero. Using EIA-930 hourly data for 2025, we measured the same gap across U.S. grids:

65.5%	**124 kg**	**43.7%**	**8,760**
median load matched in the same hour, across 22 U.S. balancing authorities (range 41–82%)	CO_2e per MWh consumed (median) drawn from the grid but reported as zero	median hourly match for solar-only certificates	certificates per meter per year once accounting moves to hourly matching

Hourly accounting is where the rules are heading: the GHG Protocol's Scope 2 revision consulted on it, and U.S. registries (PJM, ERCOT, CleanCounts) already issue hourly certificates. Hourly certificates need cheap issuance, a market for individual hours, claims anyone can check, and protection against bad meter data, which a blockchain alone cannot provide.

What Saat does

Step	Mechanism
1 · AI issuance verifier	Each metered hour is checked against ERA5 reanalysis weather, sun position and clear-sky irradiance by gradient-boosted median and 99th-percentile models trained on the plant's history. Rules: power after sunset; over capacity; above the weather envelope; day total or day *shape* (time shift) anomaly; 30-day drift. Held hours never enter the root. The report's SHA-256 is committed on-chain.
2 · Merkle issuance	`commitGeneration`: one root per plant per local market day (~99k gas for 24 hours on Anvil). Certificates are minted lazily as ERC-1155 tokens `id = plant ‖ hour`, 1 unit = 1 Wh, **once per plant-hour**, never above registered capacity.
3 · P2P market	`SaatMarket`: escrowed listings of one hour's certificates, so a listed hour cannot also be sold elsewhere. Scarce hours carry higher prices. Reentrancy-guarded.
4 · AI day-ahead planner	Using only the previous day's forecast (Open-Meteo previous-runs archive), gradient boosting forecasts tomorrow's certificates hour by hour. Flexible load moves into forecast-surplus hours and the shortfall is bought from the cheapest listings.
5 · Hourly retirement	The utility commits the consumer's 24 readings, and the day's full load enters the denominator at once. `retire` burns only certificates of the **same hour and zone**, capped at that hour's load (a noon certificate against 21:00 reverts with `HourMismatch`). `settleDay` does all 24 hours in one transaction. `claimSummary` returns the hourly score on-chain.



The demo day (10 July 2025): the hotel's contract alone covers 59.9% of its load hour by hour. After P2P buying and the AI load shift, the on-chain hourly score is 84.6%. **Live on Ethereum Sepolia**: SaatRegistry `0x616F…2F4A`, source verified on Sourcify; 27 transactions; `claimSummary(7, 20279, 20279)` returns 8460 bps.

Evidence on real data

AI verifier: a real 4.7 MW plant, 16 months it never saw

NREL PVDAQ system 9068 (single-axis tracker, Kersey, Colorado; DOE Open Energy Data Initiative). The verifier was trained on 2017–2022, calibrated on 2023 and tested on Jan 2024–Apr 2025, with fraud simulated on the real test data. The table shows the share of fake energy that never becomes a certificate.

Fraud	Capacity check (typical today)	Physics bound only	Saat AI
Power claimed after sunset	0%	100%	**100%**
Output relabelled 2 h later (into scarce evening hours)	0%	70%	**81%**
One hour inflated ×1.5	0%	24%	**70%**
Every hour +50%	0%	43%	**63%**
Sunny day replayed on a cloudy day	0%	14%	**48%**
Every hour +20%	0%	16%	31%
Real energy wrongly held back	0%	7.1%	**0.52%**

Hourly error is 10.4% of capacity, against 18.0% for physics alone (R^2 0.75 vs 0.42). The 30-day drift test catches 94% of persistent +30% inflations within 90 days, with no false alarms across 424 clean days. Settlement-grade meters anchor the totals; the verifier adds what metering does not see: relabelled hours, replayed days, impossible output, and self-reported rooftop and community solar.

Antalya, all of 2025

The community is a 2 MW community solar plant (Antalya) and a 6 MW Aegean wind share (Karaburun), supplying a seaside hotel and an office campus in Türkiye's zone. Supply is modelled from real ERA5 weather; the loads are synthetic; the forecasters were trained on 2024 only.

The hotel does every day what it does on the demo day. Annual accounting reports 0 t CO_2e; counted hour by hour on Türkiye's real 2025 grid, the grid power in its uncovered hours carries **281 t** a year on its contract alone and **133 t** with Saat. Its median day is 85%, so the demo day is typical. With only a quarter of the plants' spare output on offer, the community still reaches 77.6%.

The hotel, hourly matched, 2025	**%**
Annual claim	100
Contract only	62.9
+ AI day-ahead shifting	67.5
+ hourly market (18% more certificates)	**82.8**

Adoption and scale

- **Registries and grid operators.** In Türkiye, EPİAŞ already issues YEK-G guarantees of origin, one per MWh. Saat lets the issuer commit one root per plant-day from settlement data it already holds. It adds hourly granularity, local market days (UTC+3) and a public guard against double issuance and double claims.
- **Generators and energy communities** sell the hours that are worth most. Rooftop and cooperative solar without settlement-grade metering are where the AI verifier matters most.
- **Corporate buyers** (hotels, data centres, exporters facing carbon border rules) buy the hours they are short of, shift load, and publish a claim anyone can recompute from the chain.
- **Cost.** One transaction per plant per day. Minting is lazy, so only certificates that are traded or retired touch the chain. Measured gas on Anvil: commit 99k per plant-day, mint 40–50k per hour in a batch, and a consumer's whole day settles in one transaction (1.17M gas for 24 hours and 35 retirements, 30% less than hour by hour). Plain Solidity 0.8 and OpenZeppelin 5 run on any EVM chain, a low-fee rollup or a registry-run permissioned network.

Engineering

- **Contracts:** `SaatRegistry` and `SaatMarket`, with 25 Foundry tests. They include a fuzz test that matched energy never exceeds load, plus tests for tampered proofs, re-mint attempts, wrong day, wrong zone and wrong hour.
- **Python:** Merkle trees byte-compatible with Solidity, a web3.py client that signs locally (Anvil and public testnets), the verifier, the planner and the data loaders. 21 tests, including an end-to-end chain run.
- **Dashboard:** FastAPI plus Chart.js. It runs a full market day on a fresh chain with one click.
- **Reproducibility:** each study downloads its public data once and writes its results to `results/*.json`.

Data

Türkiye: EPİAŞ hourly generation and demand via the IEA Real-Time Electricity Tracker, with Ember monthly solar (both CC BY 4.0). **U.S.:** EIA-930 six-month files, 2025 (public domain). **PV meter data:** NREL PVDAQ system 9068, DOE Open Energy Data Initiative. **Weather:** ERA5 reanalysis and previous-day forecasts via Open-Meteo (CC BY 4.0). **Emission factors:** IPCC AR5 WG3 Annex III medians; oil/other per Electricity Maps defaults.

The Antalya plants are modelled from real weather, and the hotel and campus loads are synthetic; no real company is depicted. Zero-reported emissions use each region's average hourly intensity of its own generation (imports excluded). The dashboard runs the day on a local chain; `scripts/public_chain_day.py` ran the same day on Ethereum Sepolia. Code is MIT-licensed.